

NVIDIA Corporation

Recommendation **BUY** ★★★★★

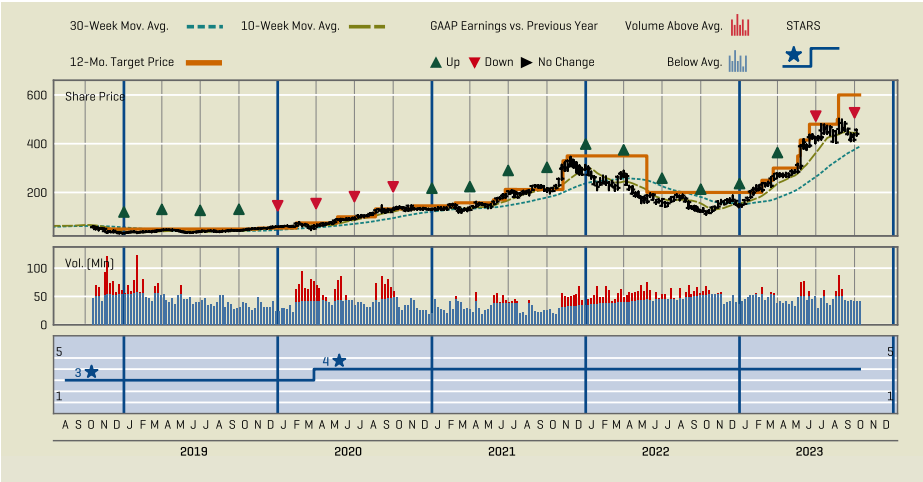
Price USD 454.61 [as of market close Oct 13, 2023] 12-Mo. Target Price USD 600.00 Report Currency USD Investment Style Large-Cap Growth

Equity Analyst Angelo Zino, CFA

GICS Sector Information Technology Summary NVIDIA develops and markets graphics and mobile processors for personal computers, Sub-Industry Semiconductors workstations, and wireless devices such as tablets and phones.

Key Stock Statistics (Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports)									
52-Wk Range	USD 502.66 - 108.13	Oper.EPS2024E	USD 10.88	Market Capitalization[B]	USD 1122.89	Beta	1.76		
Trailing 12-Month EPS	USD 5.25	Oper.EPS2025E	USD 15.44	Yield [%]	0.04	3-yr Proj. EPS CAGR[%]	130		
Trailing 12-Month P/E	86.59	P/E on Oper.EPS2024E	41.78	Dividend Rate/Share	USD 0.16	SPGMI's Quality Ranking	A-		
USD 10K Invested 5 Yrs Ago	74,404.0	Common Shares Outstg.[M]	2,489.00	Trailing 12-Month Dividend	USD 0.16	Institutional Ownership [%]	68.0		

Price Performance



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by Angelo Zino, CFA on Aug 24, 2023 08:52 AM ET, when the stock traded at USD 471.16.

Highlights

- ▶ We see sales rising 36% in FY 25 (Jan.) and 14% in FY 26 after our view for revenue to double in FY 24. Generative AI and investments in large language models are driving robust demand [data centers +171% in Jul-Q], with cloud providers increasingly adopting NVIDIA's next-gen Hopper GPUs. We believe NVIDIA's DGX on-premise/cloud offerings are seeing strong traction while its Grace CPU is now ramping with customers. Gaming rose 22% in Jul-Q (first time in five quarters), as it laps easier comps/peak demand, while automotive grew 15%. The shift to AI from traditional servers is supporting significant wallet share gains and spurring exponential growth for NVIDIA's software, GPUs, networking, and CPUs.
- ▶ We forecast a gross margin of about 71% to 72% from FY 24 to FY 26, wider than the 59% in FY 23. We see rising volume within higher-margin data centers and greater software contribution sustaining NVIDIA's elevated margins but see quarterly results somewhat dependent on end-market mix.
- ▶ We like NVIDIA's financial position, with \$16.3B in cash and \$11B in debt. We see FCF of \$24B in FY 24 and \$30B in FY 25. We believe NVIDIA is seeing improving supply from Taiwan Semi (TSM 94 ****), which is supporting surging demand.

Investment Rationale/Risk

- ▶ Our Buy view reflects our optimism surrounding growth. The emergence of generative AI is creating unprecedented demand for NVIDIA's GPUs, which are, of course, considered to be the primary building blocks for large language models and are driving permanent changes to the infrastructure of how cloud customers will operate. New software opportunities being created by NVIDIA, leveraging capabilities from the enterprise space across all industries, will only further increase demand for NVIDIA's solutions (e.g., autonomous vehicles). In addition to GPUs, NVIDIA will have the potential to leverage its new Grace CPU that it is now ramping, as well as InfiniBand networking offerings within the data center. China exposure is a risk [20%-25% sales].
- ▶ Risks to our opinion and target price include weaker-than-anticipated graphics chip sales, slower-than-expected traction for new offerings, lower spending plans from data center customers, and increased competition from other semiconductor peers.
- ▶ Our 12-month target price of \$600 is 32.6x our CY 25 operating EPS view, above peers, to reflect our view of superior growth prospects for GPUs. We note this compares to NVIDIA's five-year historical forward average of 42.2x.

Analyst's Risk Assessment

LOW	MEDIUM	HIGH
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Our risk assessment reflects industry cyclicality, the company's dependence on sales of consumer electronics, and the need to maintain an adequate amount of R&D spending to keep up with product cycles and hold market share.

Revenue/Earnings Data

Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2026	--	--	--	--	E 84,102
2025	E 16,706	E 17,951	E 19,086	E 19,799	E 73,542
2024	7,192	13,507	E 16,063	E 17,335	E 54,097
2023	8,288	6,704	5,931	6,051	26,974
2022	5,661	6,507	7,103	7,643	26,914
2021	3,080	3,866	4,726	5,003	16,675

Earnings Per Share (USD)

	1Q	2Q	3Q	4Q	Year
2026	E 4.00	E 4.40	E 4.90	E 5.10	E 18.40
2025	E 3.47	E 3.72	E 4.03	E 4.21	E 15.44
2024	1.09	2.70	E 3.34	E 3.75	E 10.88
2023	1.36	0.51	0.58	0.88	3.34
2022	0.92	1.04	1.17	1.32	4.44
2021	0.45	0.54	0.73	0.78	2.50

Fiscal Year ended Jan 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

Amount [USD]	Date Decl.	Ex-Div. Date	Stk. of Record	Payment Date
0.0400	Aug 23	Sep 06	Sep 07	Sep 28 '23
0.0400	May 24	Jun 07	Jun 08	Jun 30 '23
0.0400	Feb 22	Mar 07	Mar 08	Mar 29 '23
0.0400	Nov 16	Nov 30	Dec 01	Dec 22 '22

Dividends have been paid since 2012 . Source: Company reports
Past performance is not an indication of future performance and should not be relied as such.
Forecasts are not a reliable indicator of future performance.
Dividends paid in currencies other than the Trading currency have been accordingly converted for display purposes.



NVIDIA Corporation

Business Summary Aug 24, 2023

CORPORATE OVERVIEW: NVIDIA Corporation [NVDA], which invented the graphics processing unit [GPU] in 1999, makes products for “visual computing,” high-performance computing, and mobile computing markets.

The company has two major reporting segments: Graphics and Compute & Networking. NVDA’s Graphics segment includes GeForce GPUs for gaming and PCs, the GeForce NOW game streaming service and related infrastructure, and solutions for gaming platforms; Quadro GPUs for enterprise design; GRID software for cloud-based visual and virtual computing; and automotive platforms for infotainment systems. The Compute & Networking segment includes Data Center platforms and systems for AI, HPC, and accelerated computing; DRIVE for autonomous vehicles; and Jetson for robotics and other embedded platforms. Starting with the second quarter of fiscal 2021, Mellanox revenue is included in this segment.

NVDA’s platforms address four large markets where its expertise is critical: Gaming [34% of revenue in FY 23], Professional Visualization [6%], Data Center [56%], and Automotive [4%]. OEM and Other represented the remaining percentage of revenue.

Within gaming, products include GeForce RTX and GeForce GTX GPUs for PC gaming, SHIELD devices for gaming and streaming, GeForce NOW for cloud-based gaming, as well as platforms and development services for specialized console gaming devices. NVDA developed NVIDIA RTX, bringing next-generation graphics and AI to games. The NVIDIA RTX line-up features ray tracing technology for real-time, cinematic-quality rendering. Ray tracing, which has long been used for special effects in the movie industry, is a computationally intensive technique that simulates the physical behavior of light to achieve greater realism in computer-generated scenes.

The NVIDIA computing platform focuses on accelerating the most compute-intensive workloads, such as AI, data analytics, graphics, and scientific computing, across hyperscale, cloud, enterprise, public sector, and edge data centers. The platform consists of the company’s energy efficient GPUs, interconnects and systems, its CUDA programming model, and a growing body of software libraries, Software Development Kits, or SDKs, application frameworks, and services. In the field of AI, the company’s platform accelerates both deep learning and machine learning workloads. HPC, also referred to as scientific computing, uses numerical computational approaches to solve large and complex problems. For both AI and HPC applications, the NVIDIA accelerated computing platform greatly increases the performance and power efficiency of high-performance computers and data centers. The segment also includes Mellanox, a supplier of high-performance interconnect and networking products.

In Professional Visualization, NVDA works closely with independent software vendors to optimize their offerings for GPUs. Its GPU computing solutions enhance productivity and introduce new capabilities for critical parts of the workflow for such major industries as automotive, media and entertainment, architectural engineering, oil and gas, and medical imaging.

Automotive comprises cockpit infotainment solutions, AV platforms, and associated development agreements. Leveraging its technology leadership in AI and building on the company’s long-standing automotive relationships, it is delivering a complete end-to-end solution for the AV market under the DRIVE brand. NVIDIA has demonstrated multiple applications of AI within the car: AI can drive the car itself as a pilot in fully autonomous mode or it can also be a co-pilot, assisting the human driver while creating a safer driving experience.

IMPACT OF MAJOR DEVELOPMENTS. In September 2020, NVDA announced its intent to acquire Arm Limited for \$40 billion, which includes \$21.5 billion in stock and \$12 billion in cash, along with \$6.5 billion via meeting certain financial targets/equity to Arm employees. In February 2022, NVDA officially announced the termination of its planned acquisition of Arm Limited, citing regulatory challenges preventing deal completion. SoftBank will obtain the \$1.25 billion breakup fee and will now start preparations for a public offering, aiming for a U.S. listing in 2023. We are not surprised by the news at all, which has no impact on our target price or EPS estimates, as we have long viewed as a combination as highly unlikely to transpire given the regulatory hurdles in the U.S., the U.K., Europe, and China. In fact, we highlighted the nail in the coffin likely came in December 2021 [we assigned less than a 10% probability of deal closure] when the U.S. FTC said it was suing NVDA to block the acquisition of Arm Limited.

In April 2020, NVDA completed its acquisition of Mellanox in an all-cash transaction for \$6.9 billion, which it has available on the balance sheet, seen closing at the end of the calendar year. The deal is expected to be immediately accretive to non-GAAP gross margins, EPS, and free cash flow. We think MLNX provides greater diversification away from the volatile gaming business as it expands NVDA’s capabilities within networking and storage to address a \$60 billion-plus addressable data center market.

FINANCIAL TRENDS. NVDA’s revenues are generally cyclical and will likely trend with the economy and broader semiconductor market. Company-specific issues that can impact results include product cycles and technology. Return on assets for the 12 months in FY 23 was 8.2%, down from 17.2% in FY 22 and 12.3% in FY 21. Return on capital also declined to 9.6% in FY 23, which compares to 20.0% for FY 22 and 15.0% in FY 21. These less favorable metrics in FY 23 largely reflect a sharp industry downturn, specifically within the gaming segment. We note NVDA’s financial position remains relatively strong, with a total debt to capital ratio of 35% at the end of January.

We view NVDA as appealing for long-term investors given the potential tied to AI/data centers and self-driving vehicles. We think investors need to be forward looking for a high growth company like NVDA but note volatility to earnings given the cyclicity of the industry.

Corporate information

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Officers	
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Executive VP, General Counsel & Secretary T. S. Teter	VP & Chief Accounting Officer D. F. Robertson
Co-Founder, CEO, President & Director J. Huang	
Board Members	
A. B. Seawell	M. B. Lora
A. S. Shah	M. G. McCaffery
D. E. Hudson	M. L. Perry
H. C. Jones	P. S. Drell
J. Huang	R. K. Burgess
J. O. Dabiri	S. C. Neal
M. A. Stevens	T. Cox
Domicile Delaware	Auditor PricewaterhouseCoopers LLP
Founded 1993	
Employees 26,196	
Stockholders 344	



NVIDIA Corporation

Quantitative Evaluations						Expanded Ratio Analysis					
Fair Value Rank		1	2	3	4	5		2023	2022	2021	2020
		LOWEST						HIGHEST			
		Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued (1) to most undervalued (5).									
Fair Value Calculation		USD	Analysis of the stock's current worth, based on CFRA's proprietary quantitative model suggests that NVDA is overvalued by USD 108.13 or 23.79%								
		346.48									
Volatility		LOW		AVERAGE		HIGH					
Technical Evaluation		BULLISH	Since January, 2023, the technical indicators for NVDA have been BULLISH"								
Insider Activity		UNFAVORABLE	NEUTRAL		FAVORABLE						
Key Growth Rates and Averages											
Past Growth Rate (%)								1 Year	3 Years	5 Years	
Net Income								NM	16.03	7.47	
Sales								0.22	35.19	22.66	
Ratio Analysis (Annual Avg.)											
Net Margin [%]								16.19	26.14	27.87	
% LT Debt to Capitalization								28.43	27.08	22.44	
Return on Equity [%]								17.93	30.85	33.55	

Company Financials Fiscal year ending Jan 31											
Per Share Data (USD)		2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
Tangible Book Value		6.51	7.95	4.02	4.71	3.58	2.81	2.15	1.71	1.64	1.55
Free Cash Flow		1.53	3.26	1.90	1.75	1.29	1.21	0.69	0.50	0.36	0.25
Earnings		1.74	3.85	1.73	1.13	1.66	1.20	0.64	0.27	0.28	0.18
Earnings (Normalized)		3.34	4.44	2.50	1.45	1.66	1.23	0.76	0.42	0.36	0.25
Dividends		0.16	0.16	0.16	0.16	0.15	0.14	0.12	0.10	0.08	0.08
Payout Ratio (%)		9.11	4.09	9.12	13.95	8.96	11.19	15.67	34.69	29.48	41.14
Prices: High		289.46	346.47	147.27	64.88	73.19	60.84	29.98	8.48	5.31	4.11
Prices: Low		108.13	115.66	45.17	32.75	31.12	23.79	6.19	4.74	3.83	3.01
P/E Ratio: High		86.70	78.00	58.90	44.80	44.10	49.50	39.20	20.30	15.00	16.60
P/E Ratio: Low		32.40	26.10	18.10	22.60	18.70	19.30	8.10	11.30	10.80	12.20
Income Statement Analysis (Million USD)											
Revenue		26,974	26,914	16,675	10,918	11,716	9,714	6,910	5,010	4,682	4,130
Operating Income		5,577	10,041	4,721	2,846	3,804	3,210	1,937	878.00	759.00	496.00
Depreciation + Amortization		1,544	1,174	1,098	381.00	262.00	199.00	187.00	197.00	220.00	239.00
Interest Expense		262.00	236.00	184.00	52.00	58.00	61.00	58.00	47.00	46.00	10.00
Pretax Income		4,181	9,941	4,409	2,970	3,896	3,196	1,905	743.00	755.00	510.00
Effective Tax Rate		-4.50	1.90	1.70	5.90	-6.30	4.70	12.50	17.40	16.40	13.70
Net Income		4,368	9,752	4,332	2,796	4,141	3,047	1,666	614.00	631.00	440.00
Net Income (Normalized)		3,487	6,151	2,874	1,857	2,435	1,998	1,206	546.20	471.90	317.50
Balance Sheet and Other Financial Data (Million USD)											
Cash		13,296	21,208	11,561	10,897	7,422	7,108	6,798	5,037	4,623	4,672
Current Assets		23,073	28,829	16,055	13,690	10,557	9,255	8,536	6,053	5,713	5,625
Total Assets		41,182	44,187	28,791	17,315	13,292	11,241	9,841	7,370	7,201	7,251
Current Liabilities		6,563	4,335	3,925	1,784	1,329	1,153	1,788	2,351	896.00	945.00
Long Term Debt		9,703	10,946	5,964	1,991	1,988	1,985	1,983	7.00	1,384	1,356
Total Capital		34,132	38,443	24,490	14,847	11,330	9,471	8,545	5,903	5,816	5,830
Capital Expenditures		1,833	976.00	1,128	489.00	600.00	593.00	176.00	86.00	122.00	255.00
Cash from Operations		5,641	9,108	5,822	4,761	3,743	3,502	1,672	1,175	906.00	835.00
Current Ratio		3.52	6.65	4.09	7.67	7.94	8.03	4.77	2.57	6.38	5.95
% Long Term Debt of Capitalization		28.40	28.50	24.40	13.40	17.50	21.00	23.20	0.10	23.80	23.30
% Net Income of Revenue		16.20	36.20	26.00	25.60	35.30	31.40	24.10	12.30	13.50	10.70
% Return on Assets		8.17	17.20	12.80	11.62	19.38	19.03	14.07	7.53	6.56	4.54
% Return on Equity		17.90	44.80	29.80	26.00	49.30	46.10	32.60	13.80	14.20	9.50

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

NVIDIA Corporation

Sub-Industry Outlook

We have a positive fundamental outlook for the semiconductors sub-industry over the next 12 months. We expect semiconductor sales to rebound about 10% in 2024 after our forecast for a 12% decline in 2023, with a cyclical recovery to start in the second half of the year [Y/Y revenue growth returns in Q4 2023]. We are starting to see a return to sequential growth across the more cyclical areas of the semiconductor market (PCs, smartphones, and gaming), driven in part by seasonality as well as a more normalized inventory channel after sharply undershipping end-demand for the last several quarters. Whether or not recent strength can be sustained will be dependent on macro factors and the health of the consumer, but we think the industry is in a very healthy situation.

Currently, the biggest question mark is within the industrial/automotive markets as those areas held up well through 1H 2023, but channel inventory is now being rightsized due to end-market softness, which may continue until at least Q1 2024. Although inventories are still above optimal levels, companies are making efforts (e.g., production cuts) to quickly get back to normalized levels. In the long term, we expect content growth in automobiles to sharply outpace semiconductor industry growth, driven by several megatrends, such as autonomy, electrification, and connected infotainment. Analog semiconductors are particularly exposed to these trends.

Although data center spending has been relatively muted in 2023 (we estimate ~2%), we see a 10%-15% rebound in 2024, with investments continuing to be focused on upgrading the network to meet more artificial intelligence (AI) driven workloads. As a result, we expect accelerated computing technologies (GPUs, FPGAs, and ASICs) to gain traction, in our view, as they help overcome the limitations of CPUs. We believe that the recent spend shift towards accelerators will be permanent and will grow to an annual run rate of over \$130 billion by 2027 versus our estimated \$45 billion-\$50 billion run rate in 2023. Chipmakers tied to the data center are exposed to the best financially positioned

customers on the planet, and despite quarterly lumpiness, the long-term prospects related to themes like AI and the metaverse remain extremely encouraging and offer significant growth potential ahead.

We believe the semiconductor industry will witness a super cycle over the next decade as content growth per device across a host of markets accelerates, driven by the emergence of greater AI capabilities. Given this, we estimate the semiconductors industry will grow at a 10-year annualized pace of 6%-7% (using 2021 expected revenue as the base year), or at least 2x global GDP.

While companies that outsource manufacturing would typically have more stability in gross margins (a trade-off for capped upside), others with their own manufacturing see more variability. Industry margins are a function of manufacturing utilization and inventory supply-demand imbalances. We forecast some companies, especially in analog, to maintain higher inventory levels. Healthy free cash flow generation and high cash levels will likely result in greater shareholder return via share repurchases. Regulatory and geopolitical uncertainties remain notable risks.

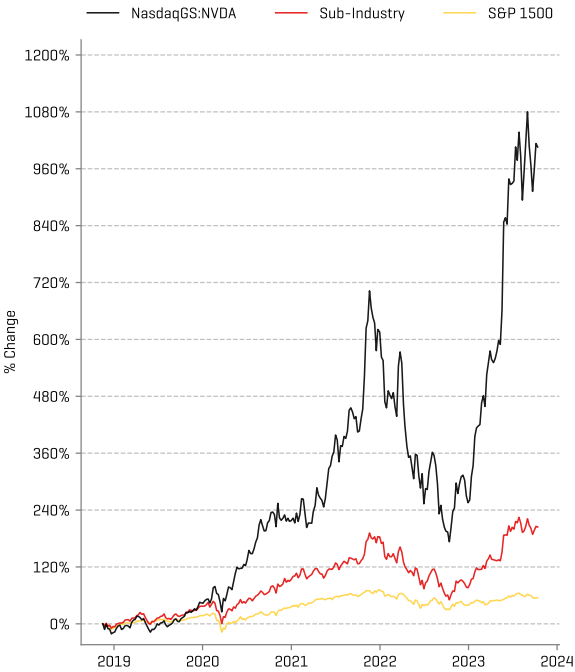
YTD through September 22, the S&P semiconductors sub-industry index rose 64.0% vs. an 11.6% increase for the S&P 1500 index. For full-year 2022, the S&P semiconductors sub-industry index fell 37.9% vs. a 19.1% decline for the S&P 1500 index.

/ Angelo Zino, CFA

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Semiconductors

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 14, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Semiconductors Peer Group*: Semiconductors

Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. (M)	30-Day Price Chg. (%)	1-Year Price Chg. (%)	P/E Ratio	Fair Value Calc.	Yield (%)	Return on Equity (%)	LTD to Cap (%)
NVIDIA Corporation	NVDA	NasdaqGS	USD	454.61	1,122,887.0	-0.0	280.1	87.0	346.48	0.0	40.2	22.0
Advanced Micro Devices, Inc.	AMD	NasdaqGS	USD	105.09	169,791.0	-2.4	78.3	41.0	N/A	N/A	-0.0	3.0
Analog Devices, Inc.	ADI	NasdaqGS	USD	171.07	85,247.0	-2.9	19.8	16.0	166.84	2.0	10.3	15.0
Arm Holdings plc	ARM	NasdaqGS	USD	50.78	52,104.0	N/A	N/A	130.0	N/A	N/A	N/A	N/A
Broadcom Inc.	AVGO	NasdaqGS	USD	883.18	364,520.0	3.6	101.6	21.0	757.20	2.1	64.8	62.2
Intel Corporation	INTC	NasdaqGS	USD	35.97	150,642.0	-7.1	36.2	46.0	N/A	1.4	-0.9	30.2
Micron Technology, Inc.	MU	NasdaqGS	USD	69.21	75,995.0	-2.3	26.2	NM	N/A	0.7	-12.4	20.5
NXP Semiconductors N.V.	NXPI	NasdaqGS	USD	194.50	50,143.0	-2.9	35.8	14.0	N/A	2.1	35.8	51.8
QUALCOMM Incorporated	QCOM	NasdaqGS	USD	108.66	121,265.0	-3.5	-3.7	11.0	143.60	2.9	47.0	40.2
Taiwan Semiconductor Manufacturing Company Limited	TSM	NYSE	USD	90.46	445,065.0	-0.8	35.8	15.0	N/A	1.7	33.8	21.8
Texas Instruments Incorporated	TXN	NasdaqGS	USD	152.75	138,692.0	-7.1	-1.0	18.0	147.30	3.4	51.2	40.2

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

NVIDIA Corporation

Analyst Research Notes and other Company News

September 15, 2023

07:22 AM ET... CFRA Sees Successful IPO of ARM Holdings as a Positive for Semiconductors [NVDA 455.81****]:

Shares of Arm Holdings [ARM 64 NR] rose about 25% on the day of its IPO, which we think demonstrates the excitement that most investors have had for AI related technology companies, specifically on the infrastructure side since the launch of ChatGPT. ARM essentially provides the building blocks for many of the leading-edge chips that are designed today, specifically on the CPU side, and makes money by licensing its IP. While ARM is seeing momentum in the data center via its Arm-based computing platforms [GPUs/CPU] and will benefit as more computing moves to the edge, it is primarily exposed to the smartphone space. ARM's jaw dropping valuation [+25x sales] for what we see as a mid-single digit percentage growth company is alarming, but we do note its highly recurring-based revenue stream/good visibility with industry leading gross margin [+90%]. We view a successful IPO boding well for semi sentiment and paving the way for more Tech IPOs. Our top AI infrastructure chip names remain AMD, AVGO, MRVL, and NVDA. / Angelo Zino, CFA

August 24, 2023

08:19 AM ET... CFRA Keeps Buy Recommendation on Shares of NVIDIA Corporation [NVDA 471.16****]:

We increase our 12-month target to \$600 from \$500, on P/E of 32.6x our CY 25 EPS view, above peers but below historical. We up our FY 24 [Jan.] EPS estimate to \$10.88 from \$8.16 and FY 25's to \$15.44 from \$12.21. We initiate FY 26 at \$18.40. After posting better-than-expected results [see prior note for details], we come away extremely bullish about NVDA's expanding TAM across all sectors/industries and increasing shift to AI from traditional servers that is spurring exponential growth for NVDA's software, GPUs, CPUs, and networking. We believe NVDA is striving given its architecture [best in class for AI inferencing], enormous installed base [attracts developers], and massive reach [e.g., cloud, edge, PC, robotics]. We like improving supply and think NVDA's earnings/FCF leverage [gross margin +70%] is underappreciated given its growth trajectory, as we see FCF growth at +\$23B in FY 24 and +\$30B in FY 25 [vs. \$3.8B in FY 23]. NVDA is growing into its multiple, now trading less than 30x our CY 25 EPS view. / Angelo Zino, CFA

August 23, 2023

05:17 PM ET... CFRA Keeps Buy Recommendation on Shares of NVIDIA Corporation [NVDA 471.16****]:

NVDA posted Jul-Q EPS of \$2.70 vs. \$0.51, beating our \$2.09 estimate. Sales rose 101% to \$13.5B, well above our view, on surging demand within its data center business [\$10.3B crushed our \$8B view], which grew 171%. Gaming also performed extremely well, up 22%, its first Y/Y increase in five quarters, as it starts to lap easy comps. We also think Oct-Q revenue guidance of \$16B, which implies growth accelerating to 170%, demonstrates that key supplier Taiwan Semi is giving NVDA preferential treatment to meet elevated demand. We believe NVDA is capitalizing on the need for cloud customers to upgrade their infrastructure, which we see as a multi-year process, and enterprise customers investing in AI initiatives for fear of being left behind. Overall, we see the accelerator market growing at a +40% annualized pace over the next three years, with NVDA taking a bulk of the wallet share. We also see massive opportunities on the software side. We will update our view after dissecting NVDA's earnings call and results. / Angelo Zino, CFA

August 21, 2023

01:23 PM ET... CFRA Keeps Buy Recommendation on Shares of NVIDIA Corporation [NVDA 452.14****]:

We boost our 12-month target to \$500 from \$480 on a P/E of ~41x our CY 24 EPS estimate, below historical, but above peers given AI prospects. We up our FY 24 [Jan.] EPS estimate to \$8.16 from \$7.85 and raise FY 25's to \$12.21 from \$10.46. Ahead of Jul-Q results, set to be released after the close on Wednesday, we look for EPS of \$2.09 [vs. \$0.51 a year ago] and sales of \$11.1B [up 65% Y/Y]. We think commentary highlighting software revenue potential [across the stack] will be key as well as guidance [Oct-Q consensus at \$2.37 EPS and sales of \$11.1B implies more than doubling Y/Y] indicating sustained demand. We see revenue led by a greater appetite for GPUs across the data center space, which we forecast at \$8B for the Jul-Q [up 110% Y/Y and 87% from Apr-Q] and growing to \$9.1B in the Oct-Q, reflecting surging orders from cloud providers and pricing power for its H100. We also see gaming returning to Y/Y growth [+15%] after four quarters of declines. We like NVDA's strong competitive moat on the software side. / Angelo Zino, CFA

June 28, 2023

07:54 AM ET... CFRA Keeps Buy Recommendation on Shares of NVIDIA Corporation [NVDA 418.76****]:

Shares of NVDA, among other chip vendors, are down pre-market on concerns the U.S. is considering new restrictions on exports of artificial intelligence chips to China. While we need to await details of the more stringent restrictions, expected to come over the next month, it will likely ban NVDA's A800 chips which were used to replace the A100 in China after certain restrictions were put in place last year [of course NVDA's newest Hopper GPUs were always banned]. Although our best assumption is that 5%-10% of revenue could be at risk [20%-25% total sales currently attached to China], demand is far outstripping supply at the moment, which would allow NVDA to absorb any potential lost China revenue by shifting sales to other vendors. That said, China remains an extremely important market for NVDA and other leading-edge chipmakers given key data center customers [e.g., Alibaba, Baidu, Tencent] in the region, so additional restrictions would inherently be a negative towards maximizing potential sales longer term. / Angelo Zino, CFA

June 16, 2023

12:41 AM ET... CFRA Maintains Buy Opinion on Shares of NVIDIA Corporation [NVDA 426.53****]:

We up our 12-month target to \$480 from \$415 on a P/E of 46x our CY 24 EPS estimate, above historical and peers given AI prospects. We keep our FY 24 [Jan.] EPS estimate at \$7.85 and raise FY 25 to \$10.46 from \$8.65. We are considerably increasing our forecast for the accelerator market [+130B by 2027 from +\$60B in our March forecast; about \$30B-\$35B seen in 2023] given commentary from cloud/enterprise players as well as from CEOs Jensen Huang and Lisa Su, demonstrating how much the landscape has changed in the last 75 days. Although we think NVDA's +90% share position in GPU data center servers is unsustainable, +25% annualized EPS growth over the next 3-5 years is more than attainable given our more attractive outlook for the broader accelerator market [GPUs to take at least 55%-60% of this share]. NVDA's closest competitor, AMD, announced details earlier this week to its MI300 accelerators that we think will gauge significant interest from cloud customers but won't deter NVDA from its own AI opportunities. / Angelo Zino, CFA

May 30, 2023

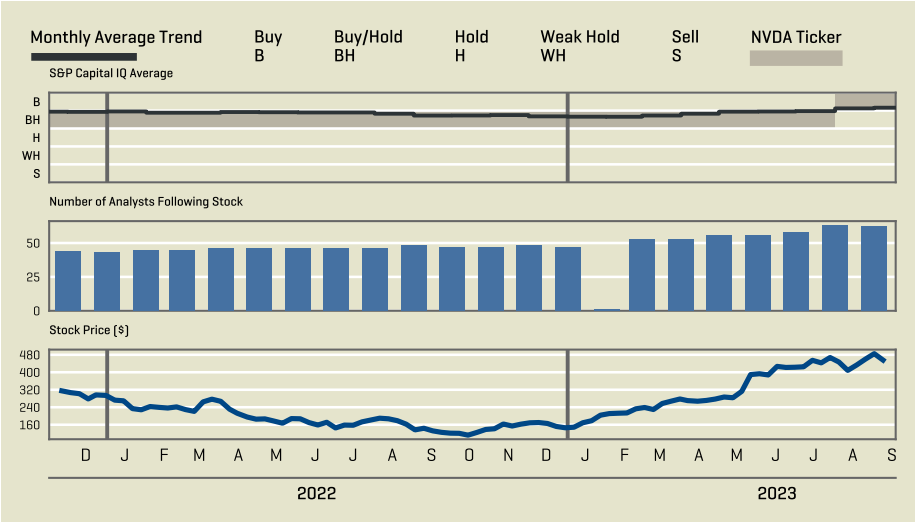
09:48 AM ET... CFRA Maintains Buy Opinion on Shares of NVIDIA Corporation [NVDA 411.74****]:

CEO Jensen Huang held a live keynote at the COMPUTEX conference in Taipei where he announced a slew of new accelerated computer offerings. Most importantly, NVDA unveiled DGX GH200, an AI supercomputer that offers 144TB of memory across 256 NVIDIA Grace Hopper Superchips [combines a Grace CPU with a Hopper GPU]. NVDA also announced offerings attached to networking technology, digital advertising, and the NVIDIA Avatar Cloud Engine [ACE] for games. We believe NVIDIA's accelerated computing GPUs stand out to be the biggest beneficiary in an AI-driven world and is the only way for cloud customers and broader enterprise space to create successful large language models at this time. Although bears will point to a daunting P/E [over 50x CY 23], we think investors need to be more forward looking for a high growth company like NVDA and note a low 40s P/E on a CY 24 basis is within the company's five-year historical forward range as well as justified given our view that EPS can grow at a five-year CAGR of +25%. / Angelo Zino, CFA

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

NVIDIA Corporation

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	38	61	38	32
Buy/Hold	12	19	12	12
Hold	3	5	3	6
Weak hold	0	0	0	1
Sell	0	0	0	0
No Opinion	9	15	9	7
Total	62	100	62	58

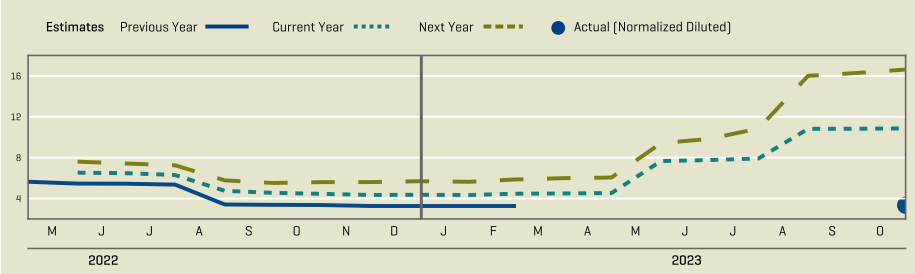
Wall Street Consensus Opinion

Buy

Wall Street Consensus vs. Performance

For fiscal year 2024, analysts estimate that NVDA will earn USD 10.86. For fiscal year 2025, analysts estimate that NVDA's earnings per share will grow by 53.15% to USD 16.63.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2025	16.63	25.62	9.71	45	27.34
2024	10.86	11.87	10.39	38	41.87
2025 vs. 2024	▲ 53%	▲ 116%	▼ -7%	▲ 18%	▼ -35%
Q3'25	4.43	6.93	3.50	31	102.68
Q3'24	3.35	3.69	3.25	38	135.58
Q3'25 vs. Q3'24	▲ 32%	▲ 88%	▲ 8%	▼ -18%	▼ -24%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

NVIDIA Corporation

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsule the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes (paid in the country of origin).

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



NVIDIA Corporation

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Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

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STARS Stock Reports:

Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

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